QPCODE Reg. No: .....

Name: .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) B Sc (Hons) GEOLOGY EXAMINATION

OCTOBER 2025

DSE - MG3DSEGEO200 - ENDOGENOUS EARTH PROCESSES

(2024 ADMISSION ONWARDS)

Duration: 1.5-hour Maximum Marks: 70 marks

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer in 60 words

Answer any 5 questions

Each question carries 2 marks

1. Mention two evidences supporting continental drift hypothesis.
2. Define the principle of Uniformitarianism.
3. What are the three main types of plate boundaries?
4. What is the geothermal gradient?
5. What is Epeirogeny.
6. Differentiate between Richter and Mercalli scales.

[2x5 = 10marks]

Part B

Short notes in 250 words

Answer any 6 questions

Each question carries 8 marks

7. Discuss the global distribution of volcanoes.
8. Discuss the classification of earthquake based on causes and depth.
9. Explain the behaviour of Magma.
10. Discuss the Isostasy concepts.
11. Explain the concept of Sea floor spreading and their supporting evidences.
12. Explain mantle convection currents and mantle plume.
13. What are endogenous geomorphic processes? Describe their associated landforms.
14. Differentiate between Himalayan, Andes, and Japan types of orogeny.

[8x6 = 48marks]

Part C

Essay in 400 words

Answer any 1 question

Each question carries 12 marks

15. Discuss the major changes that the Earth has undergone through geologic time?
16. Describe the plate tectonics theory and its significance in explaining endogenous geomorphic processes

[12x1 = 12marks]